

Al-Aqsa University
Faculty of Computers and IT
Dept. of Info & Computer science



جامعة الأقصى
كلية الحاسبات وتكنولوجيا المعلومات
قسم علم الحاسوب والمعلومات

(ITCS4315) Data Science

Assignment 2: Preprocessing & Visualization

Presented to: Eng. Nour Abo Rokba

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Assignment 2 – Submission

1. Products Dataset (Web Scraping – Assignment 1):

I collected the dataset from my own web scraping project in Assignment 1. I performed preprocessing steps including cleaning numeric columns, handling missing values, removing duplicates, normalization, and visualizations before and after. The final dataset was saved as `ayatstore_products_final.csv`.

Google Colab link:

[<https://colab.research.google.com/drive/1zgze0DXR7Xz8sbctr3dpuAjwPd8BMuD?usp=sharing>]

2. Titanic Dataset (External Dataset):

I selected the Titanic dataset because it has more than 500 rows (891 entries) and contains real preprocessing challenges such as missing values and outliers. I performed preprocessing by filling missing Age with the median, filling Embarked with the mode, dropping the Cabin column, removing duplicates, normalizing Age and Fare, and creating visualizations before and after normalization. The final dataset was saved as `titanic_final.csv`.

Google Colab link:

[https://colab.research.google.com/drive/1QVcKwcI3hW40wUFX9_2Mhpghm7JKAURf?usp=sharing]

3. Video Explanation I recorded a video explaining both datasets, the preprocessing steps, and the final results.

YouTube link: [<https://youtu.be/SK8E2ZEmzt4?si=etiH9AHwdwWEusSC>]

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